

AmritChidiya — 4-Member Team Ownership Handbook

Comprehensive Role Breakdown, Step-by-Step Engineering Execution & Member USPs (Newbie Friendly)

1. The Big Picture: How the 4 Members Work Together (For Newbies)

Think of AmritChidiya like a modern, highly secure digital government office. Each member owns one critical department:

- Member 1 (The Front Counter): Builds the welcoming touch-screen interface where the citizen talks, checks their eligibility, and receives a printed receipt to take home.
- Member 2 (The Ears & Voice): Listens to the citizen's voice in regional Indian accents, converts it into text at lightning speed, and speaks back with a human-sounding voice.
- Member 3 (The Scheme Expert & Security Detective): Knows every single government rule, guarantees 0% false promises, and checks suspicious WhatsApp links to protect citizens from scams.
- Member 4 (The Bank Vault & Cloud Guard): Locks up user passwords with unhackable 12-round Bcrypt encryption, gives out secure digital entry passes (JWT tokens), and keeps a backup database running 24/7.

2. Technical Responsibility & Deliverables Matrix

Member	Department & Role	Assigned Files	Standout Technical USP
Member 1	Frontend UI/UX & Citizen Access Lead	page.tsx CscDossierModal.tsx EligibilityCalculator.tsx MarkdownContent.tsx	Phyigital CSC Dossier Bridge: Generates physical printable receipts (AC-CSC-XXXXXX) with NPCI bank checklist for village Jan Seva Kendra centers.
Member 2	Backend API & Speech Processing Lead	main.py requirements.txt Whisper STT & Edge-TTS	Low-Latency Vernacular Audio Pipeline: High-speed Whisper Large v3 voice pipeline with Indian phonetic bias and neural audio streaming.
Member 3	AI Agent, Scheme Logic & Cyber Safety Lead	agents/chat_agent.py prompts/system_prompt.py ScamShieldModal.tsx	Anti-Hallucination Matcher & Cyber Shield: Zero-hallucination LangGraph state machine paired with domain cyber scam inspector.
Member 4	Database, Auth Security & DevOps Lead	database.py lib/auth.ts supabase_schema.sql	Zero-Trust Hybrid DB & Bcrypt Auth: Dual Supabase + SQLite failover layer with salted 12-round Bcrypt security and 24-hr JWT tokens.

3. Member 1: Frontend UI/UX & Citizen Access Lead

Phyigital Last-Mile Bridge & Citizen CSC Dossier Generator

Why this matters for a beginner: Most welfare apps fail in rural India because citizens don't own laptops or printers to apply online. Member 1 solved this by inventing the Phyigital CSC Dossier Engine: an in-app tool that generates a physical, printable receipt (AC-CSC-XXXXXX) with their verified schemes, an NPCI/DBT bank checklist, and a physical CSC VLE Operator verification stamp box. A rural citizen can simply print this sheet and walk into any village Jan Seva Kendra / Cyber Cafe to complete their application with zero confusion.

Step-by-Step Working Mechanism:

- Language Initialization: When a citizen selects Hindi or Marathi, `page.tsx` updates localized greetings, buttons, and voice models instantly.
- Instant Yojana Calculator: In `EligibilityCalculatorModal.tsx`, the citizen picks simple dropdowns (State, Category, Income, Profession). The code dynamically calculates matching welfare schemes and shows estimated annual cash benefits (e.g. ₹30,000–₹55,000/yr).
- Seamless Context Transfer: Clicking 'Get Step-by-Step Guide →' passes their profile into the chat state machine without retyping.
- Phyigital CSC Printing: In `CscDossierModal.tsx`, custom `@media print` CSS formats an A4 dossier hiding browser bars and showing official VLE sign-off boxes.
- Safe Hyperlinks: `MarkdownContent.tsx` ensures all official government links open safely in a new tab with `rel='noopener noreferrer'`.

Interview / Viva Questions & Model Answers:

Q1: What does 'Phyigital' mean and why is the CSC Dossier unique?

A: 'Phyigital' combines Physical + Digital. Most rural citizens must visit a physical Common Service Center (CSC) to complete biometric/Aadhaar verification. The CSC Dossier organizes their documents, scheme codes, and DBT checklist in advance, preventing application rejections.

Q2: How do you handle print styling in Next.js?

A: Using CSS `@media print` rules inside `CscDossierModal.tsx`. It hides the sidebar, input bar, and buttons, converting dark-mode glassmorphism into high-contrast black-and-white print ready for standard A4 printers.

4. Member 2: Backend API & Speech Processing Lead

▣ Low-Latency Vernacular Audio Pipeline with Indian Accent DSP

Why this matters for a beginner: Standard speech recognition fails on Indian accents, mixed dialects (Hinglish), and noisy environments. Member 2 created a dedicated audio pipeline using Groq Whisper Large v3 primed with custom Indian regional vocabulary hints (UP, Uttarakhand, B.Tech, OBC, scholarship, 1.5 lakh), ambient noise/silence hallucination suppression, and Microsoft Edge-TTS streaming natural human voice in 5 Indian languages.

Step-by-Step Working Mechanism:

1. Voice Capture & Upload: Citizen speaks $\rightarrow$ browser records a WebM audio blob $\rightarrow$ POSTed to `/transcribe` endpoint in `backend/main.py`.
2. Regional Phonetic Prompting: Member 2 supplies Whisper with an exact prompt of Indian states, categories, and education levels so Whisper doesn't mistranslate Indian terms into English.
3. Silence Hallucination Cleansing: Whisper often hallucinates phrases like 'Thank you for watching' when someone pauses. Member 2's code checks an exact blacklist of silence artifacts and strips them cleanly.
4. Text-to-Speech Sanitization: The AI response has markdown asterisks (`**bold**`), hashtags (`###`), and links. Member 2's `clean_text_for_speech` function removes all markup so Edge-TTS reads smooth, lifelike sentences.
5. FastAPI Routing & Security: Hosts `/chat`, `/transcribe`, `/tts`, `/signup`, `/login`, enforces CORS origin regex (`^https://.*\.vercel\.app$`), and blocks DDoS attacks with SlowAPI.

Interview / Viva Questions & Model Answers:

Q1: Why did you choose Groq Whisper over other Speech-to-Text APIs?

A: Groq runs on specialized LPU (Language Processing Unit) chips, providing sub-500ms transcription latency. This makes voice conversations feel real-time rather than having an awkward 3-second delay.

Q2: How does the text sanitization function work before TTS?

A: `clean_text_for_speech()` uses regular expressions to strip out markdown asterisks, hashes, code blocks, and URLs. It turns bullet points into natural sentence pauses so the voice sounds like a caring human companion.

5. Member 3: AI Agent, Scheme Logic & Cyber Safety Lead

▣ Anti-Hallucination Welfare Match Engine & Domain Cyber Shield

Why this matters for a beginner: If an AI makes up a fake scholarship or sends a student to an expired link, citizens lose faith in the system. Member 3 built a strict LangGraph state machine that prohibits fake schemes, combined with the **Official Domain Shield & Scam Checker** (`ScamShieldModal.tsx`) that verifies authentic `.gov.in` domains and protects citizens from fake WhatsApp schemes demanding registration fees.

Step-by-Step Working Mechanism:

1. Phase 1 (Profile Intake): If a student says 'Mujhe scholarship chahiye', the agent checks if State, Income, Category, and Education are provided. If missing, it asks polite clarifying questions and suppresses scheme extraction.
2. Phase 2 (Eligibility Match): Once details are verified, the agent outputs exact scheme names formatted in structured markdown headers.
3. Government URL Mapper: `SCHEME_URL_MAP` in `chat_agent.py` maps each scheme name directly to official government portals (`scholarships.gov.in`, `pmkisan.gov.in`, `scholarship.up.gov.in`).
4. Phase 3 (Step-by-Step Roadmap): Delivers a 5-step registration breakdown with required document checklists.
5. Cyber Shield Inspection: When a citizen pastes a WhatsApp link into `ScamShieldModal.tsx`, the code extracts the hostname, checks for official `.gov.in`/`.nic.in` accreditation, and flags phishing TLDs (`.xyz`, `.top`).

Interview / Viva Questions & Model Answers:

Q1: How does LangGraph prevent LLM hallucinations in government schemes?

A: LangGraph enforces a structured state machine. The prompt explicitly prohibits suggesting schemes during profile intake, and the regex extractor only accepts schemes formatted in strict headers that match verified criteria.

Q2: How does the Cyber Shield protect against WhatsApp fraud?

A: It inspects the domain structure. All genuine Indian central and state welfare portals must end in `.gov.in` or `.nic.in`. If a link ends in `.xyz` or asks for private UPI fees, it triggers a high-risk scam alert.

6. Member 4: Database, Auth Security & DevOps Lead

🔒 Zero-Trust Hybrid Cloud Persistence with Bcrypt Salted Auth

Why this matters for a beginner: User passwords must be protected with bank-grade security, and a welfare app must never crash if the internet or cloud goes down. Member 4 built a dual-database architecture: connects to Supabase Cloud PostgreSQL when online, with an automatic, instant fallback to local SQLite without crashing the user session. Hardened security with salted Bcrypt (12 rounds), signed 24-hour HS256 JWT tokens, and SlowAPI rate limiting.

Step-by-Step Working Mechanism:

1. Bcrypt Password Salting: When a user signs up, Member 4's `_hash_password()` generates a unique 12-round cryptographic salt. Even if two users have the same password, their hashes are completely different.
2. Digital Wristbands (JWT Tokens): Upon login, the backend issues an HS256 signed JWT token valid for 24 hours. The frontend attaches this token in the `Authorization: Bearer` header for all private chat calls.`
3. Dual-Database Failover: `database.py` tries Supabase Cloud PostgreSQL first. If offline, it routes queries to local SQLite (amrit_chidiya.db`) without throwing an unhandled exception.`
4. Rate Limiting (SlowAPI): Limits requests (5/min for signup, 10/min for login, 30/min for chat) to prevent bots from crashing the server.
5. DevOps & Production Deployment: Manages environment variables on Render (Backend) and Vercel (Frontend), ensuring cross-cloud CORS works seamlessly.

Interview / Viva Questions & Model Answers:

Q1: Why is Bcrypt safer than plain SHA-256 for passwords?

A: SHA-256 is designed to be fast, which makes it easy for attackers with GPUs to try billions of guesses per second. Bcrypt is intentionally slow and salted, making brute-force and rainbow table attacks computationally unfeasible.

Q2: Why use a hybrid Supabase + SQLite database?

A: Supabase Cloud allows users to access their saved chats from any phone or computer. But if an internet drop occurs in a rural village, the automatic SQLite fallback guarantees the app continues working locally.